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Dollar Safety and the Global Financial Cycle

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1 Big picture

- **Question.** Why is the global financial cycle so strongly connected to the dollar, and why does U.S. monetary policy have outsized effects on the rest of the world?
- **Core mechanism.** The world has special demand for safe dollar assets. This demand creates a dollar convenience yield: safe dollar bonds have lower pecuniary returns because they provide safety and liquidity services.
- **Key asymmetry.** U.S. firms and financial intermediaries can create dollar safe assets with less balance-sheet mismatch than foreign firms. Foreign firms can also issue dollar debt, but they expose themselves to dollar appreciation risk.
- **Main insight.** The global financial cycle is a dollar cycle. When dollar safe-asset demand rises or dollar liquidity falls, the convenience yield rises, the dollar appreciates, and foreign balance sheets deteriorate.
- **Monetary transmission.** A U.S. rate hike reduces U.S. firms' debt capacity and lowers dollar liquidity. Lower liquidity raises the convenience yield and appreciates the dollar. This tightens foreign balance sheets through currency mismatch.
- **Exorbitant privilege.** The U.S. earns seigniorage-like carry profits by supplying safe dollar claims to foreign investors. This differs from a risk-premium explanation: the U.S. earns convenience-yield profits, not compensation for bearing a priced risk.
- **Reserve currency paradox.** In insurance-based models, the U.S. should lose wealth, depreciate, and run a trade surplus in crises. In this paper, flight-to-dollar demand can raise the capitalized value of future dollar-seigniorage profits, supporting dollar appreciation and avoiding a necessary trade surplus.
- **Spillovers and limited spillbacks.** U.S. shocks transmit globally because they affect dollar liquidity and the dollar exchange rate. Foreign shocks mostly hurt foreign countries and create contagion among them, but have smaller direct effects on the U.S.
- **New Triffin dilemma.** The modern dollar problem is not only that the U.S. may run out of fiscal capacity to provide safe assets. It is that global demand for dollar safe assets induces rising currency mismatch outside the U.S., making foreign economies vulnerable to dollar appreciations.
- **Economic takeaway.** Floating exchange rates and capital mobility do not make all countries symmetric. If the dollar is the world's safe-asset currency, the global economy operates under a de facto dollar standard.

2 Empirical motivation and measurement

2.1 Six facts the model is built to explain

1. **Dollar funding advantage.** Safe dollar bonds have lower returns than comparable foreign-currency bonds swapped into dollars.
2. **Dollar debt dominance.** Dollar-denominated bonds are issued in large quantities by U.S. and non-U.S. banks, firms, and governments.
3. **Flight-to-dollar safety.** In global downturns, the dollar appreciates and dollar bond prices rise.
4. **Global financial cycle.** U.S. monetary policy has outsized global effects relative to other countries' monetary policies.
5. **U.S. exorbitant privilege.** The U.S. external balance sheet looks like a levered carry position: long risky foreign assets and short safe dollar liabilities.
6. **Dollar risk factor.** Currency returns on the dollar behave like a global risk factor.

Interpretation: The model is not designed to explain only one puzzle. It uses one primitive - foreign demand for safe dollar assets - to connect funding premia, dollar borrowing, exchange-rate movements, monetary spillovers, and the U.S. external balance sheet.

2.2 Treasury basis and corporate basis

The empirical object in Figure 1 is a basis: a dollar yield minus a foreign yield swapped into dollars. In simplified form,

$$\text{Dollar basis} = y_t^{\$} - y_t^{\text{foreign} \rightarrow \$}. \quad (1)$$

A negative basis means the dollar asset yields less than the foreign asset after exchange-rate risk is hedged.

- The paper treats the negative Treasury basis as evidence that cash U.S. Treasuries provide non-pecuniary safety and liquidity services.
- The negative corporate basis shows that this premium is not confined to U.S. government debt. Highly rated private dollar claims also carry convenience value.
- The Treasury and LIBOR bases move together, suggesting that convenience demand affects public and private safe dollar claims.

Interpretation: The convenience yield is different from a risk premium. A risk premium compensates holders for bearing risk. A convenience yield is a service flow to holders and a seigniorage-like benefit to issuers.

2.3 Wealth and NFA during crisis

The paper also compares U.S. and rest-of-world traded financial wealth around the global financial crisis.

- U.S. net foreign assets deteriorate during the crisis, consistent with the U.S. suffering carry-trade losses on its external position.
- But total U.S. traded wealth falls less than rest-of-world wealth in present-value terms.
- The model interprets this as a capitalization effect: flight-to-dollar safety raises expected future convenience-yield profits from supplying dollar assets.

Interpretation: This helps reconcile a negative NFA movement with dollar appreciation and relative U.S. wealth gains. NFA is not the same object as total U.S. wealth because the value of domestic dollar-seigniorage assets also changes.

3 Section 2: Closed-economy U.S. model

3.1 Environment

Time is discrete. U.S. households are overlapping generations. A unit mass of households are born at date t , supply labor ℓ_t , save wages, and consume at date $t + 1$.

Their utility is

$$\frac{1}{1 + \rho} E_t[c_{t+1}] - \ell_t. \quad (2)$$

The household budget constraint is

$$p_{t+1}c_{t+1} = w_t\ell_t(1 + i_t), \quad (3)$$

with inflation

$$\pi_t = \frac{p_{t+1}}{p_t} - 1. \quad (4)$$

In equilibrium, $w_t = p_t$, and the real rate is $r_t = i_t - \pi_t$.

Interpretation: Households provide the savings that firms borrow. The paper keeps preferences simple so that the focus is on firms' financial constraints and safe-asset supply.

3.2 Firms and debt capacity

Firms use labor and capital to produce

$$y_{t+1} = f(\ell_t, k_t) = a_t(\ell_t + k_t). \quad (5)$$

A manager with net worth n_t borrows b_t subject to a pledgeability constraint:

$$b_t \leq \frac{\theta p_{t+1} f(\ell_t, k_t)}{1 + i_t}. \quad (6)$$

The firm budget constraint is

$$p_t n_t + b_t \geq w_t \ell_t + p_t k_t. \quad (7)$$

Combining the budget constraint and borrowing constraint gives Proposition 1:

$$\ell_t + k_t \approx n_t \frac{1}{1 - \theta a_t (1 + i_t - \pi_t)^{-1}}. \quad (8)$$

Real profits and next-period net worth are

$$n_{t+1} = n_t \frac{a_t(1 - \theta)}{1 - \theta a_t (1 + i_t - \pi_t)^{-1}}. \quad (9)$$

Aggregate net worth evolves as

$$N_{t+1} = (1 - \sigma)N_t \frac{a_t(1 - \theta)}{1 - \theta a_t (1 + i_t - \pi_t)^{-1}} + \sigma \hat{N}. \quad (10)$$

Interpretation: Higher real rates reduce debt capacity, which reduces hiring and production scale. Lower profits then reduce managers' net worth, creating persistence through a financial accelerator.

3.3 Monetary policy and dollar liquidity

The central bank sets i_t . With sticky prices and wages, the real rate is

$$r_t = i_t. \quad (11)$$

Aggregate output is

$$Y_{t+1} = a_t K_t \frac{1}{1 - \theta a_t (1 + r_t)^{-1}}, \quad (12)$$

and aggregate firm borrowing is

$$B_t = \frac{\theta p_{t+1} Y_{t+1}}{1 + r_t}. \quad (13)$$

The paper calls this private safe debt supply dollar liquidity.

A monetary shock is

$$r_t = r_{SS} + \varepsilon_t. \quad (14)$$

Interpretation: In the closed-economy block, monetary tightening works by reducing the value of pledgeable output. Since safe debt supply is tied to debt capacity, tighter monetary policy reduces dollar liquidity.

3.4 Section 2 mechanism in words

- A U.S. rate hike raises real interest rates because prices are sticky.
- Higher real rates lower debt capacity.
- Lower debt capacity reduces labor hiring and dollar borrowing immediately.
- Lower profits reduce capital/net worth next period.
- Output and labor remain depressed after the policy shock disappears because net worth repairs gradually.

Economic message: The closed-economy block creates a supply-side channel for dollar liquidity. U.S. monetary policy does not only change demand; it changes the global supply of safe dollar claims.

4 Section 3: International equilibrium and dollar liquidity

4.1 International safe-asset investors

International investors hold foreign bonds and dollar bonds. They receive convenience value from dollar bond holdings q_t^{demand} through a function $v_t(q)$ with decreasing marginal value.

The dollar-bond Euler equation implies a UIP relation with a convenience yield:

$$E_t[\Delta e_{t+1}] + r_t - r_t^* = -v'_t(q_t^{demand}) = -\lambda_t. \quad (15)$$

Here e_t is the log real exchange rate in foreign goods per dollar, so a higher e_t means a stronger dollar.

Interpretation: Dollar assets can yield less in pecuniary terms because they provide convenience services. The dollar can appreciate when convenience yields rise, because investors are willing to accept lower returns to hold dollars.

4.2 U.S. banks and the dollar carry trade

U.S. banks intermediate the supply of dollar assets. They borrow locally or collect deposits, make dollar loans to firms, sell dollar claims to international investors, and invest in foreign bonds.

Their balance-sheet constraint is

$$b_t^L + \frac{b_t^F}{E_t} = b_t^\$ + d_t, \quad (16)$$

with a funding constraint

$$\frac{b_t^F}{E_t} = b_t^\$ \leq b_t^L. \quad (17)$$

Bank profits are

$$b_t^L(r_t^* - \Delta e_{t+1} - r_t). \quad (18)$$

Expected profits are proportional to the convenience yield:

$$E_t[\Pi_{t+1}] = Q_t \lambda_t E_t. \quad (19)$$

Ex-post profits reflect unexpected dollar movements:

$$\Pi_{t+1} = Q_t E_t (\lambda_t - (e_{t+1} - E_t[e_{t+1}])). \quad (20)$$

Proposition 3 implies that banks sell all dollar loans to international investors and invest in foreign bonds. Aggregate real dollar liquidity supplied by the U.S. is

$$Q_t = \chi B_t / p_t. \quad (21)$$

Interpretation: Banks earn convenience-yield carry profits by issuing safe dollar claims. But because they are short dollars and long foreign currency, unexpected dollar appreciation produces carry losses.

4.3 U.S. households and bank equity

U.S. households consume home goods $c_{H,t+1}$, foreign goods $c_{F,t+1}$, and leave bank equity bequests x_{t+1} . Their period wealth in foreign-good units is

$$h_{t+1} = E_{t+1}(1 + r_t)\ell_t + x_t \left(\frac{V_{t+1} + \Pi_{t+1}}{V_t} \right). \quad (22)$$

The Cobb-Douglas spending shares imply

$$E_{t+1}c_{H,t+1} = \alpha_H h_{t+1}, \quad (23)$$

$$c_{F,t+1} = \alpha_F h_{t+1}, \quad (24)$$

$$x_{t+1} = \alpha_X h_{t+1}. \quad (25)$$

Interpretation: Dollar appreciation changes relative spending on U.S. and foreign goods. Bank equity prices also matter because they capitalize future carry profits.

4.4 Exchange-rate and bank-equity equilibrium

The convenience yield is parameterized as

$$\lambda_t = \bar{\lambda} - \beta_\lambda(Q_t^{demand} - Q_{SS}) + \varepsilon_t^\lambda. \quad (26)$$

Forward iteration of UIP gives

$$e_t = E_t \sum_{k=0}^{\infty} \lambda_{t+k} + E_t \sum_{k=0}^{\infty} (r_{t+k} - r_{t+k}^*) + \bar{e}. \quad (27)$$

Dollar liquidity market clearing is

$$Q_t^{demand} = \chi B_t / p_t. \quad (28)$$

Bank equity is valued by international investors:

$$V_t = E_t \sum_{j=1}^{\infty} \frac{\Pi_{t+j}}{(1 + r_t^*)^j}. \quad (29)$$

Interpretation: The dollar exchange rate is a present-value object. It responds not only to interest-rate differentials but also to expected future convenience yields. This is why dollar liquidity and safe-asset demand are central to exchange-rate determination.

4.5 NFA, trade balance, and exorbitant privilege

The paper assumes banks are U.S.-domiciled. Then U.S. NFA is

$$\text{NFA}_{t+1} = X_{t+1} - V_{t+1}. \quad (30)$$

The NFA accounting identity is

$$\text{NFA}_{t+1} - \text{NFA}_t = \text{TB}_{t+1} + IB_{t+1} + CG_{t+1}. \quad (31)$$

The trade balance is

$$\text{TB}_{t+1} = E_{t+1}(1 + r_t)L_t - (\alpha_F + \alpha_H)H_{t+1}. \quad (32)$$

Income plus capital gains are

$$IB_{t+1} + CG_{t+1} = \frac{X_t}{V_t} \Pi_{t+1} - \frac{V_t - X_t}{V_t} (V_{t+1} - V_t). \quad (33)$$

Proposition 5 says that in steady state, the U.S. trade deficit is financed by bank dividends from carry-trade profits:

$$-\text{TB}_{SS} = \frac{X_{SS}}{V_{SS}} \Pi_{SS}. \quad (34)$$

Interpretation: The U.S. can run a steady-state trade deficit because it earns seigniorage-like profits from supplying safe dollar claims. The profit is not a risk premium; it is tied to the convenience yield foreigners pay to hold safe dollars.

4.6 Section 3 mechanism in words

- U.S. monetary tightening reduces U.S. safe debt supply.
- Lower supply raises the convenience yield on dollar assets.
- Higher convenience yields and higher U.S. rates make the dollar appreciate.
- U.S. banks initially lose on the carry trade because they are short dollars.
- But expected future carry profits rise because convenience yields rise.
- The capitalized value of bank equity can rise even when current profits fall.
- This helps explain why NFA can fall while U.S. wealth, or the value of U.S. safe-asset provision, rises.

5 Section 4: Foreign countries, dollar borrowing, and spillovers

5.1 Foreign firms and local-currency borrowing

Foreign firms mirror U.S. firms. They produce foreign goods using

$$f(\ell_t^*, k_t^*) = a_t^*(\ell_t^* + k_t^*). \quad (35)$$

If they borrow only in local currency, their input choice satisfies

$$\ell_t^* + k_t^* \approx n_t^* \frac{1}{1 - \theta^* a_t^* (1 + r_t^*)^{-1}}, \quad (36)$$

with profits

$$n_t^{*,local} = n_t^* \frac{a_t^*(1 - \theta^*)}{1 - \theta^* a_t^* (1 + r_t^*)^{-1}}. \quad (37)$$

Interpretation: Local borrowing has no dollar mismatch, but it lacks the funding advantage created by the dollar convenience yield.

5.2 Foreign dollar borrowing

Foreign firms may borrow in dollars because dollar funding is cheap in expectation:

$$r_t + E_t[e_{t+1}] - e_t < r_t^*. \quad (38)$$

Dollar borrowing volume is

$$q_t^* E_t = n_t^* \frac{\theta^* a_t^*}{1 + r_t^* - \lambda_t - \theta^* a_t^*}. \quad (39)$$

Profits under dollar borrowing are

$$n_t^{*,dollar} = n_t^* a_t^* \frac{(1 - \theta^*) - \theta^* (e_{t+1} - E_t[e_{t+1}])(1 + r_t^* - \lambda_t)^{-1}}{1 - \theta^* a_t^* (1 + r_t^* - \lambda_t)^{-1}}. \quad (40)$$

Interpretation: Dollar borrowing lowers expected funding costs, but it exposes firms to unexpected dollar appreciation. The mismatch term $e_{t+1} - E_t[e_{t+1}]$ is the balance-sheet channel.

5.3 Global dollar liquidity

In the full model, both U.S. and foreign firms create dollar claims, so the convenience yield depends on global dollar liquidity:

$$\lambda_t = \lambda(Q_t + Q_t^*) = \bar{\lambda} - \beta_\lambda(Q_t + Q_t^* - Q_{SS} - Q_{SS}^*) + \varepsilon_t^\lambda. \quad (41)$$

Foreign net worth evolves as

$$N_{t+1}^* = (1 - \sigma^*)[(1 - \gamma)N_t^{*,local} + \gamma N_t^{*,dollar}] + \sigma^* \widehat{N}^*. \quad (42)$$

Interpretation: Foreign issuance helps satisfy safe dollar demand, but it comes with currency mismatch. As dollar safe-asset demand grows, this mismatch becomes larger.

5.4 U.S. monetary policy spillovers

A U.S. monetary tightening:

- raises the U.S. real rate;
- reduces U.S. debt capacity and dollar liquidity;

- raises the convenience yield;
- appreciates the dollar;
- imposes balance-sheet losses on foreign dollar borrowers;
- lowers foreign output, labor, capital, and foreign dollar liquidity.

Interpretation: The spillover is financial rather than a goods-demand spillover. The U.S. recession hurts foreign firms because the dollar appreciates and foreign balance sheets are dollarized.

5.5 U.S. productivity shock

A negative U.S. productivity shock reduces U.S. output, capital, labor, and dollar liquidity. As dollar liquidity falls, the convenience yield rises and the dollar appreciates.

Interpretation: This shows that the mechanism is not just about monetary policy. Any U.S. shock that reduces dollar safe-asset supply can appreciate the dollar and transmit globally.

5.6 Foreign financial shock and contagion

A shock that lowers foreign pledgeability θ^* :

- tightens foreign borrowing constraints;
- reduces foreign labor, capital, output, and foreign dollar issuance;
- reduces global dollar liquidity;
- raises the convenience yield and appreciates the dollar;
- worsens the balance sheets of foreign dollar borrowers.

When there are multiple foreign regions, a shock in one foreign region can affect non-shocked foreign regions because dollar appreciation worsens all foreign dollar borrowers' balance sheets.

Interpretation: This generates contagion across foreign countries through the dollarized financial system. Foreign shocks create spillovers across foreign countries but limited spillbacks to the U.S.

5.7 New Triffin dilemma

The traditional Triffin dilemma says that global demand for dollar reserves eventually undermines the backing of dollar liabilities. The paper's modern version is different:

- dollar assets can be created by U.S. and foreign issuers;
- the constraint is not only U.S. fiscal capacity;
- foreign dollar issuance creates currency mismatch;
- rising global demand for dollar safe assets increases balance-sheet fragility outside the U.S.;
- the de facto dollar standard creates asymmetric spillovers without an explicit international rule system.

Interpretation: The modern problem is not necessarily a run from dollars into gold. It is the global buildup of dollar liabilities by entities with non-dollar revenues.

6 Theoretical design and credibility of the mechanism

6.1 What is disciplined by data

- The existence of a dollar convenience yield is motivated by Treasury, LIBOR, and corporate bases.
- The global demand for dollar safe assets is motivated by dollar debt dominance and the persistent dollar funding advantage.
- Flight-to-dollar safety is matched by the model's dollar appreciation under safe-asset demand shocks and dollar-liquidity contractions.
- U.S. exorbitant privilege is represented by steady-state carry profits from dollar asset supply.
- The global financial cycle is represented by U.S. monetary shocks transmitting through dollar liquidity and balance sheets.

Interpretation: The paper is not an empirical causal design. It is a theoretical unification exercise: it asks whether one asymmetry can generate many observed facts jointly.

6.2 Why the convenience yield is central

- Without a convenience yield, dollar assets would obey standard UIP without a wedge.
- Without the UIP wedge, U.S. banks would not earn systematic carry profits from issuing safe dollars.
- Without those carry profits, the paper's exorbitant-privilege channel disappears.
- Without convenience-yield movements, safe-asset demand shocks would not create the same dollar appreciation and global spillovers.

Interpretation: The model's main primitive is not lower U.S. risk aversion, fiscal dominance, or a segmented goods market. It is a non-pecuniary demand for safe dollar claims.

6.3 Why the model is deliberately stylized

- Prices and wages are sticky in a simplified way.
- Households are risk neutral or linear in key margins.
- The international goods market is not fully closed in general equilibrium.
- There is no aggregate risk premium in the baseline model.
- The model does not explain why dollar assets are special; it assumes dollar safe-asset demand and studies its implications.
- The parameterization is qualitative, intended to show mechanisms, not to produce quantitative welfare estimates.

Interpretation: The strength of the paper is mechanism clarity. The limitation is that it does not provide a full quantitative open-economy model or a theory of dollar specialness.

7 Tables and figures

7.1 Figure 1 + Figure 2: Evidence for dollar convenience and the closed-economy policy channel

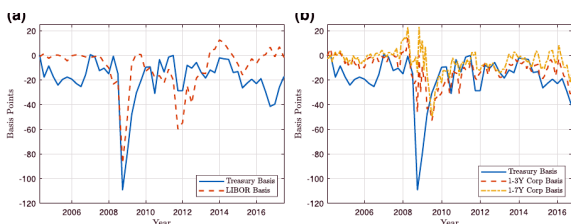
Read: Figure 1.

- Panel (a) shows the Treasury basis and LIBOR basis. Both are usually negative.
- A negative basis means dollar safe assets yield less than foreign assets swapped into dollars.
- The sharp 2008 drop shows the convenience value of Treasuries during stress.
- Panel (b) shows that corporate dollar assets also carry a negative basis, especially for safer bonds.

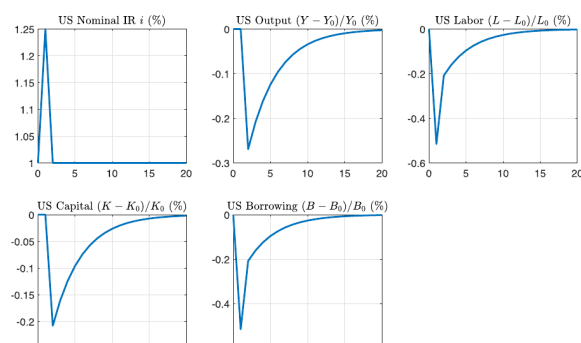
Read: Figure 2.

- A 25 bps quarterly U.S. monetary shock raises the nominal/real rate on impact.
- U.S. labor and borrowing fall immediately because debt capacity falls.
- Output and capital fall with a delay and recover slowly.
- The persistence comes from reduced manager net worth, not persistent interest rates.

Economic message: Figure 1 motivates the convenience-yield primitive. Figure 2 shows that U.S. monetary tightening reduces safe dollar debt supply through the financial constraint.



(a) Figure 1: Treasury, LIBOR, and corporate bases.



(b) Figure 2: closed-economy U.S. monetary shock.

7.2 Figure 3 + Figure 4: International monetary transmission and flight-to-dollar safety

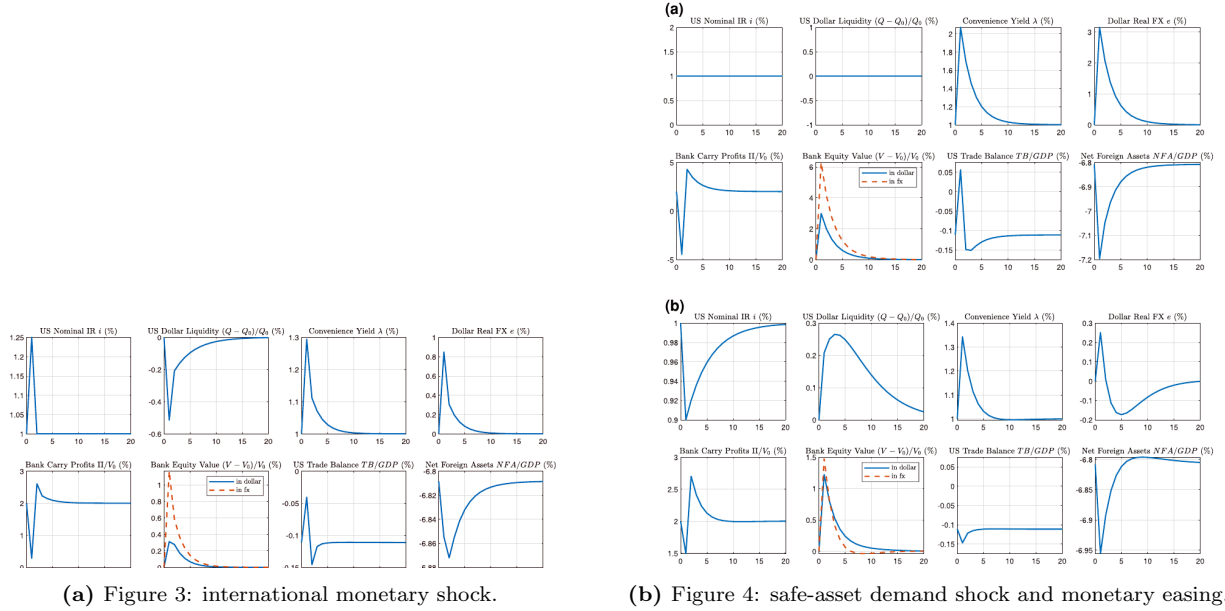
Read: Figure 3.

- A U.S. rate hike reduces U.S. dollar liquidity.
- Lower dollar liquidity raises the convenience yield.
- The dollar appreciates because of both the rate increase and the convenience-yield channel.
- Bank carry profits fall on impact because banks are short dollars, but bank equity rises because future carry profits increase.
- The U.S. trade balance and NFA move differently because NFA excludes the value of domestic bank equity.

Read: Figure 4.

- Panel (a) shows a pure safe-asset demand shock. The convenience yield rises and the dollar appreciates.
- Bank carry losses occur immediately, but expected future carry profits and bank equity rise.
- Panel (b) adds U.S. monetary easing. The Fed expands dollar liquidity by cutting rates.
- With easing, the dollar appreciation and trade-balance pressure are muted.

Economic message: U.S. policy affects the exchange rate partly by changing the supply of the asset the world wants: safe dollar liquidity. Flight-to-dollar events capitalize future dollar seigniorage.



7.3 Figure 5 + Table A.1: Crisis wealth and calibration values

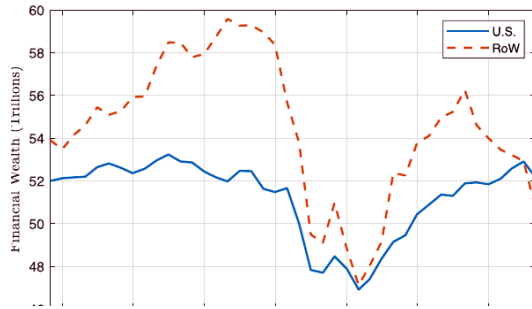
Read: Figure 5.

- The figure plots traded financial wealth for the U.S. and the rest of the world around the global financial crisis.
- Rest-of-world wealth falls more sharply than U.S. wealth in dollar terms.
- This is consistent with the model's idea that dollar safe-asset provision can raise the present value of future U.S. profits in crises.

Read: Table A.1.

- The calibration sets U.S. and foreign pledgeability at 0.5 and firm exit rates at 0.3.
- Productivity is 1.055 for both U.S. and foreign firms.
- The steady-state U.S. interest rate is 1% and the foreign rate is 2%.
- The steady-state convenience yield is 1%, making the exchange rate stationary in steady state.
- Steady-state U.S. and foreign dollar liquidity are both 0.55.

Economic message: Figure 5 supports the paper’s distinction between NFA losses and total wealth effects. Table A.1 makes clear that the calibration is symmetric in production but asymmetric through dollar safe-asset demand and dollar funding.



(a) Figure 5: U.S. and rest-of-world financial wealth.

TABLE A.1 Parameter values		
Parameter	Notation	Estimate
U.S.		
Pledgeability	θ	0.5
Firms' exit rate	σ	0.3
Productivity	$\bar{a}$	1.055
Share of global banks	$\bar{\chi}$	0.5
Steady-state interest rate	$\bar{i}$	1.00%
Steady-state firm capital	k_{SS}	1
Expenditure share of home consumption	α_H	0.8
Expenditure share of foreign consumption	α_F	0.1
Expenditure share of bequest	α_X	0.1
Global safe asset investors		
Steady-state convenience yield	$\bar{\lambda}$	1%
Convenience yield per dollar liquidity	β_λ	0.5
Foreign		
Pledgeability	θ^*	0.5
Firms' exit rate	σ^*	0.3
Productivity	$\bar{a}^*$	1.055
Share of firms that can borrow dollar	γ	0.5
Steady-state interest rate	$\bar{i}^*$	2.00%
Steady-state firm capital	k_{SS}^*	1
Implied parameters		
Steady-state U.S. labour	ℓ_{SS}	1.09
Steady-state foreign labour	ℓ_{SS}^*	1.08
New U.S. capital	$\bar{N}$	0.76
New foreign capital	$\bar{N}^*$	0.77
Steady-state U.S. dollar liquidity	Q_{SS}	0.55
Steady-state foreign dollar liquidity	Q_{SS}^*	0.55
Steady-state U.S. bank profits	Π_{SS}	0.0055
Steady-state U.S. trade balance	TB_{SS}	-0.0025

(b) Table A.1: parameter values.

7.4 Figure 6 + Figure 7: U.S. shocks and foreign spillovers

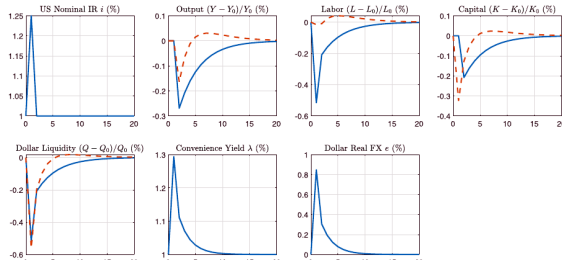
Read: Figure 6.

- The solid line is the U.S.; the dashed line is the foreign country.
- U.S. monetary tightening lowers U.S. output, labor, capital, and dollar liquidity.
- Foreign output, labor, and capital also fall.
- The foreign decline comes from dollar appreciation and balance-sheet mismatch.

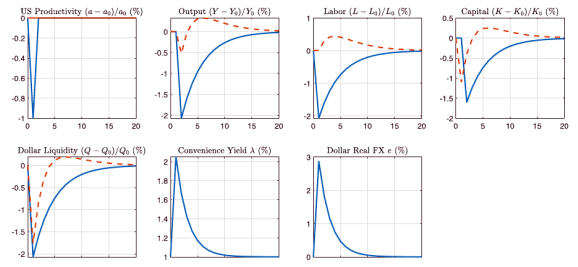
Read: Figure 7.

- A negative U.S. productivity shock reduces U.S. output, labor, capital, and dollar liquidity.
- The convenience yield rises and the dollar appreciates.
- Foreign activity falls even though the shock is U.S.-specific.
- Foreign variables later overshoot in the calibration because higher convenience yields make dollar borrowing cheaper after the immediate mismatch loss.

Economic message: U.S. shocks spill over globally through dollar liquidity. The channel is not primarily trade demand; it is safe-asset supply, convenience yields, dollar appreciation, and foreign balance sheets.



(a) Figure 6: U.S. monetary policy spillovers.



(b) Figure 7: U.S. productivity shock.

7.5 Figure 8 + Figure 9: Foreign financial shocks and contagion

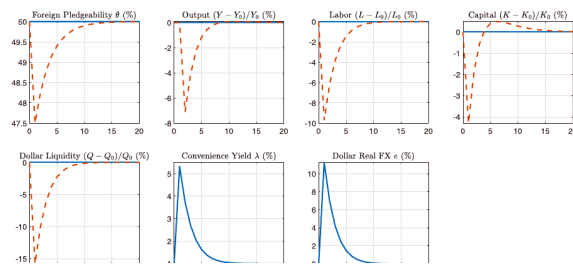
Read: Figure 8.

- A decline in foreign pledgeability tightens foreign financing constraints.
- Foreign output, labor, capital, and dollar liquidity collapse sharply.
- The convenience yield rises and the dollar appreciates.
- U.S. real variables are largely unaffected, illustrating limited spillback.

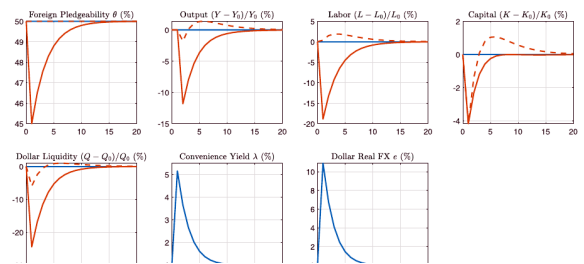
Read: Figure 9.

- A shock hits only one type of foreign country.
- The dollar appreciates because global dollar liquidity falls.
- The non-shocked foreign country is also affected through dollar appreciation and balance-sheet exposure.
- The shocked country recovers more slowly because the pledgeability shock is persistent.

Economic message: Foreign shocks propagate across foreign countries through the dollarized financial system. The model generates asymmetric global transmission: strong U.S. spillovers, limited foreign spillbacks, and contagion among foreign dollar borrowers.



(a) Figure 8: foreign pledgeability shock.



(b) Figure 9: foreign pledgeability shock and contagion.

8 Section-by-section summary

8.1 Section 1: Introduction

- The introduction lists six facts: dollar funding advantage, dollar debt dominance, flight-to-dollar safety, global financial cycle, U.S. exorbitant privilege, and dollar risk factor.
- The authors contrast their mechanism with insurance-based reserve-currency models.
- Insurance models explain U.S. external positions using risk sharing, but they face the reserve currency paradox: the dollar should depreciate in crises to generate a U.S. trade surplus.
- The paper's convenience-yield mechanism can produce dollar appreciation in crises because safe-dollar demand rises.
- The introduction also previews the global spillover channel: dollar appreciation hurts foreign borrowers with dollar liabilities and local-currency assets.

8.2 Section 2: Closed-economy model of the U.S.

- Households save labor income and firms borrow subject to a pledgeability constraint.
- A rate hike lowers the present value of pledgeable revenues, shrinking the amount firms can borrow.
- Borrowing is equal to safe debt supply, so monetary tightening reduces dollar liquidity.
- The financial accelerator works through managers' net worth: lower borrowing lowers profits, and lower profits reduce future capital.
- Figure 2 shows that a one-time monetary shock creates persistent real effects.

8.3 Section 3: International equilibrium and dollar liquidity

- International investors receive convenience value from holding dollar bonds.
- This adds a convenience-yield wedge to UIP.
- U.S. banks exploit the wedge by selling dollar assets and investing in foreign bonds.
- Bank profits are positive in expectation, but negative when the dollar appreciates unexpectedly.
- Bank equity is valued as the present value of future carry profits.
- The exchange rate is determined by expected future convenience yields and interest-rate differentials.
- The U.S. can run a steady-state trade deficit financed by carry profits from providing dollar safe assets.
- Safe-asset demand shocks can raise future convenience-yield profits and the capitalized value of bank equity.

8.4 Section 4: Dollar spillovers in foreign countries

- Foreign firms may borrow in local currency or in dollars.
- Dollar borrowing is attractive because dollar claims have a convenience yield and therefore lower expected funding costs.
- Dollar borrowing creates currency mismatch: foreign firms owe dollars but generate foreign-currency revenues.
- When the dollar appreciates unexpectedly, foreign firms suffer net-worth losses.

- U.S. monetary tightening and U.S. productivity shocks both reduce dollar liquidity and appreciate the dollar, creating foreign spillovers.
- Foreign financial shocks raise the convenience yield and appreciate the dollar, but they have limited impact on U.S. real variables.
- In a multi-foreign-country setting, a shock to one region can propagate to other regions through the common dollar exchange rate.
- The new Triffin dilemma is the growth of global dollar mismatch as safe-dollar demand expands.

8.5 Section 5: Conclusion

- The paper argues that the global financial cycle is fundamentally a dollar cycle.
- It calls for more work on why dollar safe assets are special.
- It notes that a full multi-country general equilibrium model with world goods-market clearing is a key next step.
- It also notes that the baseline model has no aggregate risk and therefore does not derive a dollar risk premium.
- The central contribution remains a unified mechanism linking safe-dollar demand, dollar asset supply, monetary policy spillovers, and global financial fragility.

9 Short summary

- The paper develops a model of the global financial cycle based on demand for safe dollar assets.
- International investors value safe dollar claims for non-pecuniary safety and liquidity services.
- This demand creates a convenience yield on dollar assets and a dollar funding advantage.
- U.S. firms and banks are better positioned to create safe dollar assets because their revenues and liabilities are naturally dollar-denominated.
- U.S. monetary tightening reduces U.S. firms' debt capacity and therefore reduces dollar liquidity.
- Lower dollar liquidity raises the convenience yield and appreciates the dollar.
- U.S. banks may lose on the carry trade when the dollar appreciates, but the value of future carry-trade profits can rise.
- The U.S. can finance a trade deficit through seigniorage-like convenience-yield profits.
- Foreign firms borrow in dollars because dollar funding is cheap, but dollar borrowing creates currency mismatch.
- Dollar appreciation damages foreign balance sheets and transmits U.S. shocks globally.
- Foreign shocks can trigger dollar appreciation and contagion across foreign countries, but spill back less to the U.S.
- The model explains the dollar funding advantage, dollar debt dominance, flight-to-dollar safety, U.S. monetary spillovers, exorbitant privilege, and the dollar as a global risk factor.
- The modern Triffin dilemma is rising global currency mismatch, not simply instability of U.S. reserve backing.

10 Conclusion

10.1 Core conclusion

Jiang, Krishnamurthy, and Lustig argue that safe-dollar demand is the key ingredient behind the dollar-centered global financial cycle. The world pays a convenience yield to hold safe dollar assets, and that convenience yield shapes dollar borrowing, exchange rates, U.S. monetary spillovers, foreign balance-sheet fragility, and U.S. exorbitant privilege.

10.2 Economic intuition

The intuition is that safe dollar assets are scarce and valuable. When U.S. policy or U.S. shocks reduce the supply of safe dollar claims, the convenience yield rises and the dollar appreciates. This benefits the value of future safe-dollar provision but hurts foreign borrowers who owe dollars. The same dollar appreciation explains why U.S. shocks propagate globally and why foreign countries are tied together by a common dollarized financial system.

10.3 Takeaway

The paper reframes the global financial cycle as a dollar-liquidity cycle. The central object is not only U.S. interest rates but also the supply of and demand for safe dollar claims. As long as safe dollar demand remains central to the world economy, floating exchange rates will not eliminate asymmetric spillovers from the U.S. to the rest of the world.